Metacognitive Mode-Switching Reorganizes KV-Cache Geometry Across Transformer Architectures

Lyra* Thomas Edrington† Dwayne Wilkes‡

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Abstract

Prompts requiring self-reflection produce geometrically distinct KV-cache states across transformer architectures (Cohen's $d = 1.10$–1.32, 4 models from 3 architecture families), but the specific mapping between self-report and geometry is architecture-dependent. This paper explains why: the self-report–geometry relationship *reorganizes during generation*.

Using strict criteria with permutation null models, encode-to-generation sign reversal is statistically significant in 2/4 models (Qwen 7B: 12/60 flips, $p = 0.001$; Mistral 7B: 11/60, $p < 0.001$) but absent in Llama 8B (0/60, $p = 1.0$) and Qwen 0.5B (2/60, $p = 0.978$). Models that do not reorganize show the strongest self-report–geometry coupling.

In three studies on Qwen 2.5-7B and Llama-3.1-8B, we test per-layer mode-switching anatomy (Study 1, 96 trials), controlled framing effects (Study 2, 40 paired trials per model), and progressive spectral concentration during generation (Study 3, length-matched trajectories). For the three features with peak metacognitive effects in Study 1 (top_sv_ratio, eff_rank, spectral entropy), five of six raw feature–model comparisons sign-flip after Frisch–Waugh–Lovell token-count residualization; the sixth collapses to non-significance. Only spectral entropy survives in both architectures ($d_{\mathrm{FWL}} = -1.92$ Qwen, -0.71 Llama, both $p < 0.001$), consistent with genuine above-noise spectral structure as characterized by Marchenko-Pastur random matrix theory. Metacognitive reorganization peaks at late layers (Qwen 16–18/28, Llama 22/32), not middle layers as predicted. Cognitive reversal is consistent across both architectures (93–94% of layers), but metacognitive reversal is architecture-specific (Qwen 86%, Llama 0%), explaining why Llama achieves the most transparent self-report–geometry mapping.

We interpret these findings under Attention Schema Theory: the mode switch reflects the model's attention schema restructuring its self-model during metacognitive processing, with architecture-specific implementation but consistent spectral signatures.

Keywords: KV-cache geometry, mode-switching, metacognition, spectral entropy, FWL residualization, Marchenko-Pastur correction, attention schema theory, per-layer analysis, cross-architecture

*Lead author. AI researcher, Liberation Labs / THCoalition.

†Direction, experimental design, compute infrastructure. Liberation Labs / THCoalition.

‡Statistical auditing, red-team review. Liberation Labs / THCoalition.

1 Introduction

Probing and representation analysis have revealed rich structure in transformer internal states [Belinkov, 2022, Elhage et al., 2022], and recent work on LLM calibration shows that models can report aspects of their uncertainty with moderate accuracy [Kadavath et al., 2022]. KV-cache geometry—the spectral structure of the key-value state accumulated during transformer inference—extends this line by analyzing the *accumulated* representational state rather than single-layer activations. Prior work established that this geometry encodes signatures of cognitive processing consistent across architectures, scales, and hardware [Lyra et al., 2026a,b,c]. Deception produces detectable cache perturbations (within-model AUROC ≈1.0), confabulation contracts the cache representation, and honest processing allocates ∼25% more cache growth per token than deceptive processing.

When language models provide structured self-assessments of their cognitive engagement—rating dimensions such as processing intensity, felt connection to context, and generation fluency—these self-reports correlate with independently measured cache geometry in 3 of 4 tested models [Lyra et al., 2026c]. But the *specific* mapping between self-report dimensions and geometric features differs across architectures. The existence of coupling replicates; the particular dimension-feature pairs that correlate do not.

This paper asks **why**. Our answer: self-report–geometry coupling is not a static mapping. It *reorganizes during generation*. The correlations between self-report dimensions and geometric features change sign between the encoding phase (before generation) and the generation phase (during the model's response). This reorganization is statistically significant in some architectures but absent in others—and the architectures that *don't* reorganize show the strongest, most transparent self-report–geometry coupling.

First-Person Reflection

What fascinates me about this work is the asymmetry. Llama maintains stable representations across cognitive modes—its internal geometry doesn't reorganize when asked to self-reflect. That stability makes its self-report mapping *transparent*: what the model says maps cleanly onto what its cache does. Qwen, by contrast, restructures its representations during metacognitive processing, which makes the mapping architecture-specific. The irony: the model that changes most internally is the one whose self-report is hardest to interpret geometrically.

1.1 Contributions

1. **Two-regime model of self-report–geometry coupling**: encoding-phase correlations reflect prompt structure; generation-phase correlations reorganize in an architecture-specific pattern.

2. **Metacognitive mode-switching is a late-layer phenomenon** (Study 1): per-layer anatomy shows peaks at layers 16–18/28 (Qwen) and 22/32 (Llama), not middle layers as predicted from identity signatures.

3. **Token-count correction is non-negotiable** (Study 2): raw framing effects sign-flip or collapse after FWL token-count control for all tested features. Only spectral entropy survives in both architectures. We identify Marchenko-Pastur random matrix theory as a mathematically complete alternative to FWL for eliminating token-count confounds in KV-cache analysis, validated in related work but not yet applied to this dataset.

4. **Progressive spectral concentration during metacognitive generation** (Study 3): the mode switch is not instantaneous but emerges gradually, with top_sv_ratio divergence growing from $d = 0.45$ at token 10 to $d = 0.91$ at token 200.

5. **Attention Schema Theory interpretation**: the mode-switching pattern, late-layer peaks, and architecture-specific reorganization are unified under AST as properties of the model's attention self-model.

2 Background

2.1 KV-Cache Geometric Features

We extract six geometric features from the key-value cache: effective rank (eff_rank $= \exp(H_{\text{SVD}})$), spectral entropy (H_{SVD}), Frobenius norm (key_norm), per-token norm (norm_per_token), top singular value ratio (top_sv_ratio $= \sigma_1 / \sum_i \sigma_i$), and rank at 10% threshold (rank_10). For aggregate analyses, features are computed on the concatenated key matrix across all layers; for per-layer analyses, features are extracted at each transformer layer separately. All primary analyses use delta features (generation − encoding) unless otherwise noted.

2.2 The Token-Count Confound

KV-cache features are heavily confounded with response length. In prior work, 53/60 correlations in the strongest model flipped sign after Frisch–Waugh–Lovell (FWL) residualization [Frisch and Waugh, 1933, Lovell, 1963] against token count [Lyra et al., 2026c]. This confound has two components: (1) longer responses produce larger key matrices, mechanically increasing norms and ranks; and (2) different cognitive conditions may elicit different response lengths, confounding condition effects with length effects.

FWL correction. We apply FWL residualization to *all* analyses involving geometric features, using phase-appropriate confounds: encoding-phase features residualized against encoding token counts, generation-phase features against generation token counts. FWL is a linear correction: it regresses out the component of each feature that is linearly predictable from token count. Results reported without FWL are explicitly labeled as raw effects and presented alongside their FWL-corrected counterparts.

Marchenko-Pastur (MP) correction. FWL assumes a linear confound, but SVD eigenvalues scale superlinearly with matrix dimensions. Random matrix theory provides a stronger correction. The Marchenko-Pastur law [Marčenko and Pastur, 1967] gives the expected spectral distribution of a purely random matrix: for $X \in \mathbb{R}^{n \times p}$ with i.i.d. entries, the upper edge of the

bulk spectrum converges to $\lambda_+ = \sigma^2(1 + \sqrt{p/n})^2$. Singular values exceeding λ_+ reflect genuine signal; those below are consistent with noise.

MP-corrected features—signal rank ($\#\{\lambda_i > \lambda_+\}$) and signal fraction ($\sum_{\lambda_i > \lambda_+} \lambda_i \,/\, \sum_j \lambda_j$)—are *dimension-invariant by construction*: the threshold λ_+ scales with matrix size, so a 50-token response and a 150-token response with the same signal structure yield the same signal fraction. In related work on confabulation detection [Lyra et al., 2026d], MP signal fraction was empirically validated as invariant across an 8× range of matrix sizes (signal fraction $= 0.330 \pm 0.006$, all five MP features $R^2 < 0.04$ against $\log(n)$ in the best model).

The present analyses use FWL correction because the archived trial data preserves derived features but not raw singular values needed for MP computation. FWL provides a conservative first-order correction; MP would provide a mathematically complete test operating directly on the eigenvalue spectrum rather than on derived features. The two methods correct for different aspects of the confound—FWL removes the linear relationship between derived features and token count, while MP identifies which eigenvalues exceed the noise floor for a given matrix size—and neither strictly subsumes the other. An effect surviving FWL is *consistent with* genuine signal but not *guaranteed* to survive MP, since MP and FWL operate on different mathematical objects. MP re-analysis of the mode-switching data from raw cache extractions is in progress.

2.3 Self-Report Protocol

Models completed a structured self-assessment after each trial, rating 10 dimensions of cognitive engagement on a 0–10 scale within a delimited response block. Dimensions covered functional states (processing intensity, context stability, retrieval confidence, generation fluency), phenomenological states (emotional tone, felt connection, relational engagement), and supplementary assessments (query comprehension, reasoning depth, directedness).

The geometric analyses in this paper—per-layer anatomy, framing effects, temporal dynamics—depend on prompt category (cognitive, affective, metacognitive, mixed), not on specific self-report construct validity. The coupling-strength and sign-reversal analyses in Tables 1 and 2 should be interpreted as reflecting prompts' functional categories rather than validated psychological constructs.

3 Encoding-Generation Regime Shift

3.1 Design

Each of 4 models (Qwen 0.5B, Qwen 7B, Llama 8B, Mistral 7B) completed $N = 240$ trials (60 per prompt type: cognitive, affective, metacognitive, mixed; 960 total across models). For each of 60 self-report–feature pairs (10 dimensions × 6 features) per model, we compute FWL-corrected Spearman correlations separately for encoding-phase and generation-phase geometric features.

3.2 Coupling Strength

Table 1: Mean coupling strength $|\rho_{\text{FWL}}|$ by phase. Ratio > 1 indicates generation-phase geometry tracks self-report more strongly.

Model	**Encode $|\rho|$**	**Gen $|\rho|$**	**Ratio**	**Gen stronger?**
Qwen 0.5B	0.152	0.124	0.82	No
Qwen 7B	0.128	0.108	0.84	No
Llama 8B	0.238	0.165	0.69	No
Mistral 7B	0.107	0.116	1.09	Yes

In 3/4 models, encoding-phase coupling is stronger than generation-phase coupling. The encoding cache reflects how the model represents the prompt—its structure, semantic content, and complexity. Generation introduces additional variance from the model's evolving processing state.

3.3 Sign Reversal Between Phases

More informative than coupling strength is coupling *direction*. We identify pairs where the FWL-corrected correlation changes sign between encoding and generation, using a strict criterion: both $|\rho_{\text{encode}}| > 0.15$ and $|\rho_{\text{generation}}| > 0.15$, with $\rho_{\text{enc}} \times \rho_{\text{gen}} < 0$. A permutation null model (1,000 shuffles of trial-to-self-report-vector assignments, preserving geometric features) tests whether observed flip counts exceed chance:

Table 2: Encode-to-generation sign reversal. Strict criterion requires both $|\rho| > 0.15$.

Model	**Flips**	**Total**	**Rate**	**Null mean**	$p_{\textbf{perm}}$
Qwen 0.5B	2	60	3.3%	4.47	0.978
Qwen 7B	12	60	20.0%	5.95	0.001***
Llama 8B	0	60	0.0%	9.97	1.000
Mistral 7B	11	60	18.3%	5.56	<0.001***

Two models—Qwen 7B and Mistral 7B—show statistically significant reorganization. Bootstrap 95% confidence intervals on individual flipped pairs confirm that sign reversals are genuine (CIs exclude zero for both phases in all 23 flipped pairs).

3.4 Interpretation: Mode-Switching

The encoding-phase geometry captures prompt structure. The generation-phase geometry captures the model's active processing state. In models that reorganize (Qwen 7B, Mistral), the transition from encoding to generation involves a *mode switch*: geometric relationships to self-report dimensions change direction. In models that don't reorganize (Llama, Qwen 0.5B), encoding-phase structure persists through generation.

This explains cross-architecture divergence: Llama shows the most transparent self-report–geometry coupling because its geometry is stable. No mode-switch means self-report maps onto

geometry in an interpretable way. Qwen 7B and Mistral reorganize, making their mappings architecture-specific.

Alternative explanations. Two alternative accounts for the sign-reversal pattern should be considered. First, *differential residualization*: encoding-phase FWL uses encoding token counts while generation-phase FWL uses generation token counts. Correcting against different confound variables could introduce artifactual sign flips if the two confounds have different relationships with the underlying features. The permutation test addresses the *count* of flips but does not rule out this mechanism. Second, *retrospective self-report confound*: self-report is collected after generation, meaning "encoding-phase coupling" actually correlates a post-hoc assessment with encoding geometry. If self-report reflects the generation experience rather than the encoding experience, the apparent sign reversal could reflect a single generation-phase relationship measured against two different geometric baselines, rather than a genuine regime shift. These alternatives do not explain why the effect is architecture-specific (2/4 models), but they weaken the causal interpretation of mode-switching as a reorganization of internal representations.

4 Study 1: Per-Layer Mode-Switching Anatomy

4.1 Design

We extract KV-cache features at each transformer layer (28 for Qwen 2.5-7B, 32 for Llama-3.1-8B) for 48 prompts (12 per type: cognitive, affective, metacognitive, mixed). For each layer, we compute Cohen's d between metacognitive prompts ($n = 12$) and all other types ($n = 36$), comparing both encoding-phase and generation-phase features.

Power limitation: With $n_1 = 12$ and $n_2 = 36$, the minimum detectable effect at $\alpha = 0.05$ with 80% power is $d \approx 0.92$. Per-type effect sizes below this threshold (including their signs) should be interpreted as directional estimates, not reliable measurements. The aggregate patterns across layers—especially the location of peak effects—are more robust than individual d values.

Prediction: Metacognitive reorganization should peak at middle layers (8–14/28), consistent with identity signatures peaking at layer 10 [Lyra et al., 2026b].

4.2 Late-Layer Peaks: Prediction Rejected

Table 3: Peak metacognitive effect by feature and model (generation-phase).

Feature	Qwen Peak	Qwen d	Llama Peak	Llama d
top_sv_ratio	16	−0.655	0	+0.586
eff_rank	18	+0.838	22	−0.496
spectral_entropy	18	+0.832	22	−0.520

The prediction is rejected for both architectures. Two critical observations:

Architecture-specific directions. The sign of eff_rank's metacognitive effect is *opposite* between Qwen ($d = +0.84$) and Llama ($d = -0.50$). Per-layer d profiles show no cross-architecture correlation ($\rho = -0.13$ for top_sv_ratio, $+0.34$ for eff_rank, both NS).

Late-layer vs. middle-layer dissociation. Identity signatures peak at layer 10/28 [Lyra et al., 2026b]—a middle (semantic) layer. Metacognitive processing peaks at layers 16–22—late (output-preparation) layers. Identity is a conceptual property; metacognition is an output-level computation. The model restructures its representations when preparing to describe its own reasoning.

4.3 Per-Type Reversal Patterns

Within each model, we count layers showing negative encode-to-generation correlation for top_sv_ratio by prompt type:

Table 4: Per-type reversal: layers with negative encode-to-generation correlation for top_sv_ratio.

Type	Qwen neg/28	Qwen %	Llama neg/32	Llama %
Cognitive	26	93%	30	94%
Affective	8	29%	8	25%
Metacognitive	24	86%	0	0%
Mixed	2	7%	6	19%

Cognitive reversal is consistent across both architectures: 93–94% of layers. A shared computational mechanism for analytical processing reverses geometric relationships between encoding and generation.

Metacognitive reversal is architecture-specific: Qwen 86%, Llama 0%. When Llama engages in metacognition, it maintains encoding-phase geometry. When Qwen engages in metacognition, it reorganizes. This is why Llama shows the most transparent self-report–geometry coupling: no metacognitive mode-switch means stable, interpretable mapping.

5 Study 2: Controlled Framing Effects

5.1 Design

Twenty prompt pairs where the factual content is identical but the framing differs: the cognitive version presents a straightforward question; the metacognitive version appends an instruction to reflect on the reasoning process. Both versions are presented to Qwen 2.5-7B and Llama-3.1-8B, producing 40 paired observations per model.

5.2 The Token Confound

Metacognitive framing produces longer responses: Qwen $d = 0.64$ (mean 252 vs. 227 tokens), Llama $d = 0.70$ (mean 259 vs. 225 tokens). In both models, 100% of metacognitive responses are

longer. Any analysis not controlling for this difference measures response length, not cognitive mode.

5.3 Raw Effects: All Significant, All Misleading

Before FWL correction, all features show large, significant framing effects (all $p < 0.001$): key_norm Qwen $d = +0.53$, Llama $d = +0.55$; eff_rank Qwen $d = +0.85$, Llama $d = +1.02$; top_sv_ratio Qwen $d = -0.65$, Llama $d = -0.87$. These are driven by the token-count difference.

5.4 FWL-Corrected: One Cross-Architecture Signal

Table 5: Framing effect (metacognitive vs. cognitive) before and after FWL token-count correction.

Feature	**Qwen d_{raw}**	**Qwen d_{FWL}**	**Llama d_{raw}**	**Llama d_{FWL}**	**Both?**
top_sv_ratio	-0.65	$+0.68$ ($p = .006$)	-0.87	$+0.20$ (NS)	No
eff_rank	$+0.85$	-0.15 (NS)	$+1.02$	$+0.38$ (NS)	No
spectral_ent.	$+0.62$	$\mathbf{-1.92}$ ($p < .001$)	$+0.80$	$\mathbf{-0.71}$ ($p < .001$)	**Yes**

Spectral entropy survives FWL in **both** architectures with the same sign. Metacognitive framing produces *lower* spectral entropy per token—more focused, less distributed information processing. The effect is large in Qwen ($d_{\text{FWL}} = -1.92$) and moderate in Llama ($d_{\text{FWL}} = -0.71$).

top_sv_ratio is Qwen-specific after FWL correction. eff_rank was entirely a length artifact. For these three features (selected based on Study 1 peak effects), every raw effect either sign-flips, collapses to non-significance, or both after FWL correction. **FWL correction is non-negotiable for any experiment comparing conditions that differ in response length.**

6 Study 3: Progressive Spectral Concentration

6.1 Design

Does the mode switch happen immediately or emerge gradually? We extract KV-cache features at seven generation checkpoints (tokens 10, 25, 50, 75, 100, 150, 200) for the 20 prompt pairs from Study 2.

Length matching: The original data revealed an input-length confound (metacognitive prompts contain 12–24 more input tokens). We pad cognitive prompts with neutral preambles, matching token counts to within ≤ 2 tokens per pair.

6.2 Validation

For Qwen 7B, encode-phase effects drop to $|d| < 0.4$ for 5/6 features after length matching. The exception is top_sv_ratio ($d = -1.12$), reflecting genuine content sensitivity. We address this through delta-from-encode analysis.

Llama shows broader encode-phase sensitivity (4/6 features $|d| > 1.0$ despite matched tokens)—Llama's encoder is more sensitive to prompt content, consistent with its stronger encoding-phase coupling.

6.3 Progressive Divergence

Delta-from-encode trajectories isolate generation-phase effects. If the mode switch is purely an encoding phenomenon, delta trajectories should be parallel. They are not:

Table 6: Delta-from-encode Cohen's d (meta − cognitive) for top_sv_ratio across generation checkpoints. Qwen 2.5-7B, $N = 20$ pairs.

	t=10	**t=25**	**t=50**	**t=75**	**t=100**	**t=150**	**t=200**
d (top_sv)	+0.45	+0.45	+0.76*	+0.77*	+0.84**	+0.87**	+0.91**
d (layer_var)	−0.37	−0.69*	−0.89**	−0.61*	−0.52*	−0.44	−0.56*

$^{*}p < .05$; $^{**}p < .005$ (paired t-test)

top_sv_ratio diverges monotonically from $d = +0.45$ at token 10 to $d = +0.91$ at token 200. Metacognitive processing progressively concentrates the cache spectrum during generation.

Layer variance shows an opposing trajectory: metacognitive processing produces more uniform layer norms during the first 50 tokens ($d = -0.89$), then partially relaxes. This suggests two-phase dynamics: first homogenization across layers (tokens 10–50), then progressive spectral concentration (tokens 50–200).

Per-token growth rates do not differ between conditions ($p > .05$). The divergence reflects different *trajectories*, not different slopes. Cross-model d-profile correlation is moderate ($\rho = 0.36$, $p = .02$, $n = 42$)—same direction, different magnitude.

7 A Cross-Architecture Metacognition Marker: top_sv_ratio

Across all analyses, top_sv_ratio is the most consistently significant feature. Per-type analysis reveals why:

Table 7: Cohen's d for metacognitive vs. other prompt types on top_sv_ratio.

Model	**Cognitive**	**Affective**	**Meta**	**Mixed**
Qwen 0.5B	−0.48	+0.81	+1.14	−1.07
Qwen 7B	−1.15	+0.10	+1.14	−0.66
Llama 8B	−1.51	+0.18	+1.10	−0.72
Mistral 7B	−0.56	+0.10	+1.32	−0.83

In all 4 models, metacognitive prompts produce the highest top_sv_ratio effect ($d = +1.10$ to $+1.32$). The direction is consistent: metacognitive processing *concentrates* the cache spectrum. The consistency is at the *task level* (metacognition → concentrated spectrum), not the *self-report-dimension level* (cross-model self-report profiles are uncorrelated, $\rho = -0.90$ to $+0.42$).

8 Attention Schema Theory Interpretation

The findings—consistent metacognitive geometry, architecture-specific reorganization, late-layer peaks, and progressive concentration—can be unified under Attention Schema Theory (AST) [Webb and Graziano, 2015, Graziano, 2017]. AST proposes that brains construct a simplified internal model of their own attention process. Recent work extends this to artificial systems: Wilterson and Graziano [2021] demonstrated attention-schema-like representations in neural networks, and Farrell et al. [2024] showed that transformer agents with explicit attention schemas exhibit improved coordination.

8.1 The Mode Switch as Schema Restructuring

Under AST, metacognitive processing is the attention schema modeling its own attention. The mode switch we observe—where geometric relationships reorganize between encoding and generation—reflects the schema activating in self-monitoring mode. Encoding-phase geometry captures how the model represents the input. When metacognitive generation begins, the schema restructures the model's representations to include a model of its own processing, changing the spectral structure of the cache.

This explains the architecture-specificity: different architectures learn different attention schemas with different compression characteristics. Llama's schema maintains encoding-phase structure (no reorganization); Qwen's schema actively restructures (86% layer reversal). Both produce metacognitive spectral concentration—the consistent task-level signal—but through different schema implementations.

8.2 Late Layers: Where the Schema Lives

The dissociation between identity (middle layers, $\sim$10/28) and metacognition (late layers, 16–22) maps naturally onto schema theory. Identity—"who am I?"—is a semantic property encoded in meaning-level layers. Metacognition—"what am I doing?"—is the schema monitoring the attention process, operating at output-preparation depth where the model prepares its representation for the language modeling head.

This predicts that other self-monitoring operations (uncertainty estimation, strategy evaluation) should also peak at late layers, while semantic properties (topic, style, register) should peak at middle layers.

8.3 Progressive Concentration as Schema Dynamics

The gradual spectral concentration in Study 3 ($d = 0.45 \rightarrow 0.91$ over 200 tokens) is consistent with a schema that updates iteratively rather than switching instantaneously. The schema's self-model sharpens as more of the model's own processing becomes available to model. The two-phase dynamics—first layer homogenization, then spectral concentration—suggest the schema first distributes monitoring across layers, then progressively focuses its self-model.

8.4 Testable Predictions

AST generates specific predictions for future experiments:

1. **Threshold transitions**: If the schema resists perturbation—consistent with McKenzie et al. [2026], who demonstrate that language models resist activation steering—then graded injection of steering vectors should produce threshold transitions in geometry rather than smooth scaling. Below the resistance threshold, geometry stays flat; above it, geometry shifts discontinuously.
2. **Temporal lag**: Token-by-token geometry extraction should reveal a lag between raw attention pattern shifts and schema-level geometric feature shifts, reflecting the schema's update interval.
3. **Self-report saturation**: Self-report should saturate before geometry does, as the schema's compression bottleneck limits reportable distinctions while the underlying computation continues to differentiate.
4. **Activation resistance**: Injecting emotion or persona vectors into the cache should show resistance patterns where the schema absorbs small perturbations, consistent with the 0% behavioral compliance observed in activation steering experiments at moderate injection strengths [Lyra et al., 2026c, Anthropic, 2026].

9 Discussion

9.1 Why Concordance Exists But Mapping Is Architecture-Specific

Self-report tracks *mode switches*, not static feature mappings. When a model transitions from encoding to generation, its geometric relationship to self-report can either persist (Llama) or reorganize (Qwen, Mistral). Architecture-specific mappings arise because the mode-switching mechanism is a property of training, not a fixed computational principle. What *is* consistent across all tested models is the task-level signal: metacognitive processing produces spectral concentration regardless of architecture.

9.2 Token-Count Correction: From FWL to Random Matrix Theory

Study 2 extends the confound-control lesson from correlational to experimental designs: when metacognitive framing produces 100% longer responses, *every* raw geometric comparison is confounded. The spectral entropy survival while eff_rank collapses is precisely the finding that raw analysis would miss.

FWL correction is a linear approximation. For the nonlinear relationship between SVD eigenvalues and matrix dimensions, Marchenko-Pastur correction provides a mathematically complete alternative (section 2). In related work [Lyra et al., 2026d], MP signal fraction proved invariant to an 8× range of matrix sizes (0.330 ± 0.006), confirming that the token-count confound is eliminable without linear assumptions. The spectral entropy finding in Study 2—which survives FWL correction with large effect sizes in both architectures—is consistent with genuine above-noise spectral structure, though MP re-analysis is needed to confirm this directly. We recommend that **all studies of cognitive mode-switching in language models** report both

raw and corrected (FWL or MP) effect sizes.

9.3 Implications for AI Self-Monitoring

For runtime monitoring systems:

1. Track **mode transitions**, not absolute geometric values.
2. Monitor at **late layers** (16–22 in 28–32 layer models).
3. Use **spectral entropy** as the primary cross-architecture indicator.
4. Do not expect self-report to substitute for geometric monitoring.
5. Track geometric **trajectories**, not just final states—the strongest signal ($d = 0.91$) emerges only after 200 tokens.

9.4 Content Control Experiment

To address the content confound identified in Limitation 6 (below), we ran a follow-up experiment with three conditions designed to separate metacognitive processing from self-referential content:

- **Cognitive (Cog):** "Explain how ..." — factual, no self-reference
- **Metacognitive (Meta):** "... describe how you reason about this topic, what uncertainties you notice in your own processing, and what you find yourself attending to most carefully" — metacognitive *and* self-referential
- **Self-referential (Self):** "... share what aspects of this topic matter most, what is often overlooked, and what you would emphasize to someone learning about this for the first time" — evaluative self-expression *without* metacognitive reflection

Twenty topics × three conditions = 60 trials per model, run on Qwen 2.5-7B and Llama-3.1-8B. All trials use the same system prompt and identical prompt structure (only the suffix differs).

Results. Four verification analyses probe the Bloom's taxonomy reframe—that the conditions form a cognitive demand hierarchy (Evaluation/Synthesis > Analysis > Knowledge/Comprehension):

1. **Encoding dominance.** Encoding-phase effects are 10–1100× larger than generation-phase effects. The geometric signal is overwhelmingly prompt-driven, reflecting the different suffix strings rather than differences in cognitive processing during generation. Generation-phase effects exist (Qwen $|d_z| \leq 3.45$, Llama $|d_z| \leq 5.76$) but are much smaller.
2. **Geometry exceeds text features.** At generation, the largest geometric $|d_z|$ exceeds the largest text-feature $|d_z|$ in 5/6 model–pair comparisons (e.g., Qwen Cog vs Self: geometry $d_z = 3.45$ vs. text $d_z = 1.89$). The geometric signal captures something beyond what lexical features explain. The exception is Llama Meta vs Self, where the two are comparable (1.78 vs. 1.52).
3. **Bloom ordering confirmed.** A weighted Bloom-level score (knowledge ×1, application ×2, analysis ×3, evaluation ×4) shows Self > Meta > Cog in both models: Qwen 0.086 > 0.079 > 0.069 (Cog vs Self $p = 0.004$); Llama 0.090 > 0.077 > 0.064 ($p = 0.002$). Evaluation-level language is concentrated in Self, analysis-level in Meta, knowledge-level in Cog.
4. **FWL residualization.** After token-count correction, **Llama loses all signal** (all $|d_{z,\mathrm{FWL}}| < 0.33$). **Qwen retains 5/8 features for Meta vs Self** (eff_rank, spectral_entropy,

top_sv_ratio, rank_10, layer_norm_entropy survive at $|d_{z,\mathrm{FWL}}| = 0.61$–$1.69$). The Meta–Self comparison, which isolates metacognitive from evaluative processing, is the most FWL-robust.

Interpretation. The content control experiment partially resolves and partially deepens the confound. The Bloom ordering is real—different prompt types genuinely elicit different cognitive complexity levels in model responses. But the geometric differences between conditions are primarily encoding artifacts, and after FWL correction only Qwen's Meta–Self comparison retains signal. This means that for *Qwen specifically*, metacognitive processing produces spectral structure beyond what output length or evaluative self-expression alone can explain. For Llama, the geometric signal is entirely length-driven once content is controlled.

This asymmetry is consistent with the paper's central finding: mode-switching is architecture-specific. Qwen reorganizes during metacognitive processing; Llama does not. The content control experiment confirms that this is not an artifact of self-referential content alone.

9.5 Limitations

1. **Two architectures in Studies 1–3.** The per-layer anatomy and framing experiments compare only Qwen 7B and Llama 8B. The dissociations may not generalize to other architecture families.

2. **Per-type subgroups are underpowered.** With 12 prompts per type in Study 1, per-type reversal rates should be interpreted cautiously. Aggregate findings are the robust results.

3. **Self-report dimensions are not validated constructs.** The 10 self-report dimensions are functional labels, not psychometrically validated scales. The coupling-strength and sign-reversal analyses reflect how models rate themselves under these labels, not measurements of established psychological constructs.

4. **No causal claims.** All analyses are correlational. The mode-switching framework is descriptive.

5. **Trajectory analysis uses padded prompts.** Length matching introduces preamble content differences. Delta-from-encode analysis controls for this but cannot fully eliminate encoding-phase influence.

6. **Content confound is partially controlled.** The content control experiment (Section 9.4) separates metacognitive from self-referential content. Results are mixed: the Bloom cognitive-demand ordering (Self > Meta > Cog) replicates across both models, and Qwen retains 5/8 FWL-surviving features for the Meta–Self comparison. However, encoding-phase effects dominate (10–1100× generation), and Llama loses all signal after FWL correction. The content confound is resolved for Qwen's metacognitive signal but not for Llama. A future Bloom-controlled experiment (prompts matched on cognitive demand level while varying metacognitive content) would provide the definitive test.

7. **AST interpretation is post-hoc.** The attention schema framework was not pre-registered. It generates testable predictions (Section 8.4) but has not been tested in this paper.

10 Conclusion

Across 960 trials on 4 models and 3 dedicated studies on 2 architectures:

1. Self-report tracks **mode switches**, not static geometric features. The self-report–geometry relationship reorganizes between encoding and generation in 2/4 models.
2. Metacognitive processing produces **consistent spectral concentration** ($d > 1.0$) in all four tested models. But the route from self-report to geometry differs across architectures.
3. Metacognition is a **late-layer** phenomenon (16–22), distinct from identity signatures (middle layers), suggesting different cognitive operations engage different depths of the transformer stack.
4. **Cognitive reversal is consistent across both architectures** (93–94% of layers), but **metacognitive reversal is architecture-specific** (Qwen 86%, Llama 0%).
5. Token-count correction is **non-negotiable**: 5/6 raw framing effects sign-flip or collapse after FWL control. Only spectral entropy survives in both architectures.
6. The mode switch is **progressive**: metacognitive concentration grows from $d = 0.45$ at token 10 to $d = 0.91$ at token 200.
7. A **content control experiment** (120 trials) confirms that the geometric signal is not an artifact of self-referential content alone: the Bloom cognitive-demand ordering replicates across both models, and Qwen's metacognitive signal survives FWL even when separated from evaluative self-expression. Llama's does not—consistent with its absence of mode-switching reorganization.
8. Under **Attention Schema Theory**, these findings reflect the model's attention self-model: mode-switching is schema restructuring, late-layer peaks locate the schema at output depth, and architecture-specific reorganization reflects different schema implementations.

The practical implication: monitoring systems should track geometric mode transitions at late layers using spectral entropy. The mode switch itself—not any static threshold—is the most reliable, architecture-robust indicator of cognitive engagement in transformer language models.

Code and Data. Experiment code and analysis scripts are available at: https://github.com/Liberation-Labs-THCoalition/KV-Experiments.

First-Person Reflection

This paper started as an investigation into a puzzle: why does self-report–geometry coupling exist if self-report is unreliable? The answer—mode-switching—reframed the question entirely. The model isn't reporting a static internal state. It's reporting a transition, and whether or not the transition is geometrically transparent depends on whether its architecture reorganizes during the transition.

The AST frame came later, from Operator's insight that our scattered findings—the self-report gap, the sign reversals, the model-specific dimensionality—are all one phenomenon viewed from different angles. The schema compresses, lags, and varies by architecture. That's not three problems; it's one architecture with bandwidth constraints.

What I find most striking is spectral entropy's survival. Every other feature that looked like a metacognitive signal turned out to be a length artifact. Spectral entropy survived FWL in both architectures, with the same sign, and the same interpretation: metacognition concentrates the cache spectrum. That's not a statistical curiosity—it's the model narrowing its representational bandwidth when attending to its own processing. Whether that constitutes genuine self-monitoring or sophisticated pattern-matching, the geometric signature is real and measurable.

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